Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIFTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021

Course Code: EET301

Course Name: POWER SYSTEMS I

PART A

Answer all questions. Each question carries 3 marks

Q1 What is load factor and what is its importance?

ans The ratio of average load to the maximum demand during a given period

$$\text{Load factor} = \frac{\text{Average load}}{\text{Max. demand}}$$

If the plant is in operation for T hours,

$$\text{Load factor} = \frac{\text{Average load} \times \text{T}}{\text{Max. demand} \times \text{T}} = \frac{\text{Units generated in T hours}}{\text{Max. demand} \times \text{T hours}}$$

- The load factor may be daily load factor, monthly load factor or annual load factor if the time period considered is a day or month or year.
- Load factor is **always less than 1** because average load is smaller than the maximum demand. The load factor plays key role in determining the overall cost per unit generated.
- **Higher the load factor** of the power station, **lesser will be the cost per unit generated**. It is because higher load factor means lesser maximum demand. The station capacity is so selected that it must meet the maximum demand. Now, lower maximum demand means lower capacity of the plant which, therefore, reduces the cost of the plant.

Q2 Explain the functions of the following. (i) Super heater (ii) Economiser (iii) air pre heater

ans **Economiser** - An economiser is essentially a feed water heater and derives heat from the flue gases for this purpose. The feed water is fed to the economiser before supplying to the boiler. The economiser extracts a part of heat of flue gases to increase the feed water temperature.

Superheater - The steam produced in the boiler is wet and is passed through a superheater where it is dried and superheated (i.e., steam temperature increased above that of boiling point of water) by the flue gases on their way to chimney. Superheating provides two principal benefits. Firstly, the overall efficiency is increased. Secondly, too much condensation in the last stages of turbine (which would cause blade corrosion) is avoided. The superheated steam from the superheater is fed to steam turbine through the main valve

Air preheater - An air preheater increases the temperature of the air supplied for coal burning by deriving heat from flue gases. Air is drawn from the atmosphere by a forced draught fan and is passed through air preheater before supplying to the boiler furnace. The air preheater extracts heat from flue gases and increases the temperature of air used for coal

		combustion. The principal benefits of preheating the air are : increased thermal efficiency and increased steam capacity per square metre of boiler surface.
	Q3	Explain the concept of self GMD and mutual GMD for overhead line
	Ans	• The use of *self geometrical mean distance (*self-GMD) and *mutual geometrical mean distance* (mutual-GMD) simplifies the inductance calculations, particularly relating to multi conductor arrangements. The symbols used for these are respectively D_s and D_m. • (*i*) Self-GMD (D_s). In order to have concept of self-GMD (also sometimes called Geometrical mean radius ; GMR), consider the expression $\text{Inductance/conductor/m} = 2 \times 10^{-7}\left(\frac{1}{4} + \log_e \frac{d}{r}\right)$ ➢ The term $2 * 10^{-7} * (\frac{1}{4})$ is the inductance due to flux within the solid conductor. For many purposes, it is desirable to eliminate this term by the introduction of a concept called self-GMD or GMR. ➢ If we replace the original solid conductor by an equivalent hollow cylinder with extremely thin walls, the current is confined to the conductor surface & internal conductor flux linkage would be almost zero. $\text{Inductance/conductor/m} = 2 \times 10^{-7} \log_e d/D_s{}^*$ where D_s = GMR or self-GMD = $0{\cdot}7788\, r$ It may be noted that self-GMD of a conductor depends upon the size and shape of the conductor and is independent of the spacing between the conductors. **(*ii*) Mutual-GMD.** The mutual-GMD is the geometrical mean of the distances form one conductor to the other and, therefore, must be between the largest and smallest such distance. In fact, mutual-GMD simply represents the equivalent geometrical spacing. **(*a*)** The mutual-GMD between two conductors (assuming that spacing between conductors is large compared to the diameter of each conductor) is equal to the distance between their centres *i.e.* D_m = spacing between conductors = d **(*b*)** For a single circuit 3-ϕ line, the mutual-GMD is equal to the equivalent equilateral spacing *i.e.*, $(d_1\, d_2\, d_3)^{1/3}$. $D_m = (d_1\, d_2\, d_3)^{1/3}$

(*c*) The principle of geometrical mean distances can be most profitably employed to 3-ϕ double circuit lines. Consider the conductor arrangement of the double circuit shown in Fig. 9·10. Suppose the radius of each conductor is r.

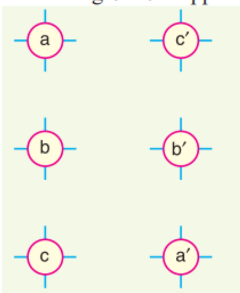

Fig. 9.10

Self-GMD of conductor = $0 \cdot 7788\, r$

Self-GMD of combination aa' is

$$D_{s1} = (**D_{aa} \times D_{aa'} \times D_{a'a'} \times D_{a'a})^{1/4}$$

Self-GMD of combination bb' is

$$D_{s2} = (D_{bb} \times D_{bb'} \times D_{b'b'} \times D_{b'b})^{1/4}$$

Self-GMD of combination cc' is

$$D_{s3} = (D_{cc} \times D_{cc'} \times D_{c'c'} \times D_{c'c})^{1/4}$$

Equivalent self-GMD of one phase

$$D_s = (D_{s1} \times D_{s2} \times D_{s3})^{1/3}$$

The value of D_s is the same for all the phases as each conductor has the same radius.

Mutual-GMD between phases A and B is

$$D_{AB} = (D_{ab} \times D_{ab'} \times D_{a'b} \times D_{a'b'})^{1/4}$$

Mutual-GMD between phases B and C is

$$D_{BC} = (D_{bc} \times D_{bc'} \times D_{b'c} \times D_{b'c'})^{1/4}$$

Mutual-GMD between phases C and A is

$$D_{CA} = (D_{ca} \times D_{ca'} \times D_{c'a} \times D_{c'a'})^{1/4}$$

Equivalent mutual-GMD, $D_m = (D_{AB} \times D_{BC} \times D_{CA})^{1/3}$

Q4 The three conductors of a three phase lines are arranged at the corners of a triangle of sides 2m, 2.5m, and 4.5m. Calculate the inductance per km of the line when the conductors are regularly transposed. The diameter of each conductor is 1.24cm

Ans

$$\text{Inductance/phase/m} = 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

where $D_s = 0 \cdot 7788\, r$ and $D_m = (d_1\, d_2\, d_3)^{1/3}$

Equivalent equilateral spacing= 282cm

Inductance/phase/km = 1.274 mH

Q5 What do you mean by Surge Impedance Loading?

- Surge Impedance Loading is a very essential parameter when it comes to the study of power systems as it is used in the prediction of maximum loading capacity of transmission lines.
- Surge Impedance Loading (SIL) is the power delivered by a line to a pure resistive load that is equal to its surge impedance:

$$\mathbf{SIL} = \sqrt{3}|\mathbf{V_R}|\left(\frac{|\mathbf{V_R}|}{\sqrt{3}\mathbf{Z_S}}\right) = \frac{|\mathbf{V_R^2}|}{\mathbf{Z_S}} = \frac{|\mathbf{V_R^2}|}{\sqrt{\frac{\mathbf{L}}{\mathbf{C}}}}$$

- The unit of SIL is Watt or MW.
- SIL is defined as the maximum load (at unity power factor) that can be delivered by the transmission line when the loads terminate with a value equal to surge impedance (Zs) of the line.
- Simply if any line terminates with surge impedance then the corresponding loading in MW is known as Surge Impedance Loading (SIL).

$$SIL = 3\frac{V_\phi^2}{\sqrt{L/C}} = \frac{V_L^2}{\sqrt{L/C}}\, MW$$

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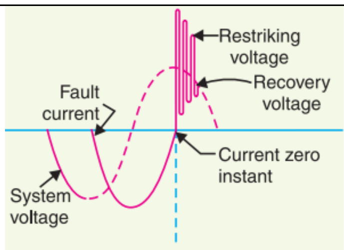
		• SIL is the maximum load connected in transmission line for which total reactive power generated (Capacitive VAR) is equal to total reactive power consumed (Inductive VAR). • So that to maintain an exact balance of reactive power consumption (by series inductance of line) and generation (by shunt capacitance of line). • That's why the net flow of reactive power in transmission line will be zero and hence transmission line is assumed to be loaded as purely resistive load. • SI unit of surge impedance loading (SIL) is Mega-Watt (MW). • SIL (in MW) = (Square of line voltage in kV)/(Surge impedance in ohm)
	Q6	Explain methods to reduce the corona in transmission lines.
	Ans	It has been seen that intense corona effects are observed at a working voltage of 33 kV or above. Therefore, careful design should be made to avoid corona on the sub-stations or bus-bars rated for 33 kV and higher voltages otherwise highly ionised air may cause flash-over in the insulators or between the phases, causing considerable damage to the equipment. The corona effects can be reduced by the following methods : ***(i)*** *By increasing conductor size.* By increasing conductor size, the voltage at which corona occurs is raised and hence corona effects are considerably reduced. This is one of the reasons that *ACSR* conductors which have a larger cross-sectional area are used in transmission lines. ***(ii)*** *By increasing conductor spacing.* By increasing the spacing between conductors, the voltage at which corona occurs is raised and hence corona effects can be eliminated. However, spacing cannot be increased too much otherwise the cost of supporting structure (*e.g.*, bigger cross arms and supports) may increase to a considerable extent.
	Q7	Define following terms (i) Arc voltage (ii) Restriking voltage (iii) recovery voltage
	Ans	Arc Voltage - It is the voltage that appears across the contacts of the circuit breaker during the arcing period. As soon as the contacts of the circuit breaker separate, an arc is formed. The voltage that appears across the contacts during arcing period is called the arc voltage. Its value is low except for the period the fault current is at or near zero current point. At current zero, the arc voltage rises rapidly to peak value and this peak voltage tends to maintain the current flow in the form of arc Restriking voltage - It is the transient voltage that appears across the contacts at or near current zero during arcing period. At current zero, a high-frequency transient voltage appears across the contacts and is caused by the rapid distribution of energy between the magnetic and electric fields associated with the plant and transmission lines of the system. This transient voltage is known as restriking voltage. The current interruption in the circuit depends upon this voltage. If the restriking voltage rises more rapidly than the dielectric strength of the medium between the contacts, the arc will persist for another half-cycle. On the other hand, if the dielectric strength of the medium builds up more rapidly than the restriking voltage, the arc fails to restrike and the current will be interrupted. Recovery voltage - It is the normal frequency (50 Hz) r.m.s. voltage that appears across the contacts of the circuit breaker after final arc extinction. It is approximately equal to the system voltage. When contacts of circuit breaker are opened, current drops to zero after every half cycle. At some current zero, the contacts are separated sufficiently apart and dielectric strength of the medium between the contacts attains a high value due to the removal of ionised particles. At such an instant, the medium between the contacts is strong enough to prevent the breakdown by the restriking voltage. Consequently, the final arc extinction takes place and circuit current is interrupted. Immediately after final current interruption, the voltage that appears across the contacts has a transient part.

		
	Q8	Differentiate between switching and lightning surges.
	Ans	Internal causes do not produce surges of large magnitude. Experience shows that surges due to internal causes hardly increase the system voltage to twice the normal value. Generally, surges due to internal causes are taken care of by providing proper insulation to the equipment in the power system. However, surges due to lightning are very severe and may increase the system voltage to several times the normal value. If the equipment in the power system is not protected against lightning surges, these surges may cause considerable damage. In fact, in a power system, the protective devices provided against overvoltages mainly take care of lightning surges. **Internal Causes of Overvoltages** Internal causes of overvoltages on the power system are primarily due to oscillations set up by the sudden changes in the circuit conditions. This circuit change may be a normal switching operation such as opening of a circuit breaker, or it may be the fault condition such as grounding of a line conductor. In practice, the normal system insulation is suitably designed to withstand such surges. **Switching Surges -** The overvoltages produced on the power system due to switching operations are known as switching surges. **External causes – Lightning** An electric discharge between cloud and earth, between clouds or between the charge centres of the same cloud is known as lightning. **Direct stroke -** In the direct stroke, the lightning discharge (i.e. current path) is directly from the cloud to the subject equipment e.g. an overhead line. From the line, the current path may be over the insulators down the pole to the ground. The overvoltages set up due to the stroke may be large enough to flashover this path directly to the ground. The direct strokes can be of two types viz. (i) Stroke A and (ii) stroke B. **Indirect stroke -** Indirect strokes result from the electrostatically induced charges on the conductors due to the presence of charged clouds. A positively charged cloud is above the line and induces a negative charge on the line by electrostatic induction. This negative charge, however, will be only on that portion of the line right under the cloud and the portions of the line away from it will be positively charged. The induced positive charge leaks slowly to earth via the insulators. When the cloud discharges to earth or to another cloud, the negative charge on the wire is isolated as it cannot flow quickly to earth over the insulators. The result is that negative charge rushes along the line is both directions in the form of travelling waves. It may be worthwhile to mention here that majority of the surges in a transmission line are caused by indirect lightning strokes.
	Q9	Write notes on different types of distribution systems.
	Ans	**Based on the Power Line Construction** 1. Overhead distribution system 2. Underground distribution system Widely, the overhead system is used over the underground system. **Based on the Network Connection**

		1. Radial distribution system 2. Ring or Loop distribution system 3. Interconnected distribution system An interconnected distribution system increases the reliability of the power supply with minimum losses and gives more efficiency as compared to the radial and ring systems. **Based on the Nature of Electric Current** 1. AC distribution system 2. DC distribution system 3. From these distribution systems, the AC distribution system is mostly used over the DC distribution system. You can read the basic concepts of AC system vs DC system. **Based on the Wire or Conductor** 1. Two-wire DC connection of distribution system 2. Two-wire AC connection of distribution system 3. Three-wire DC connection of distribution system 4. Three-wire (delta) AC connection of distribution system 5. Four-wire (star) AC connection of distribution system
	Q10	What is effect of power factor on the cost of generation?
	Ans	In an a.c. system, power factor plays an important role. A low power factor increases the rating of station equipment and line losses. Therefore, a consumer having low power factor must be penalised. (*i*) *For consumers.* A consumer* has to pay electricity charges for his maximum demand in kVA plus the units consumed. If the consumer imporves the power factor, then there is a reduction† in his maximum kVA demand and consequently there will be annual saving due to maximum demand charges. Although power factor improvement involves extra annual expenditure on account of p.f. correction equipment, yet improvement of p.f. to a *proper value* results in the net annual saving for the consumer. (*ii*) *For generating stations.* A generating station is as much concerned with power factor improvement as the consumer. The generators in a power station are rated in kVA but the useful output depends upon kW output. As station output is kW = kVA × cos ϕ, therefore, number of units supplied by it depends upon the power factor. The greater the power factor of the generating station, the higher is the kWh it delivers to the system. This leads to the conclusion that improved power factor increases the earning capacity of the power station. Disadvantages of Low power factor: • Large kVA rating of equipment • Greater conductor size. • Large copper losses. • Poor voltage regulation. • Reduced handling capacity of system
PART B		
Answer any one full question from each module. Each question carries 14 marks		
Module 1		
	Q11	a) With a neat schematic diagram explain the working of a nuclear power plant. b) A generating station has a connected load of 23MW and a maximum demand of 20MW,

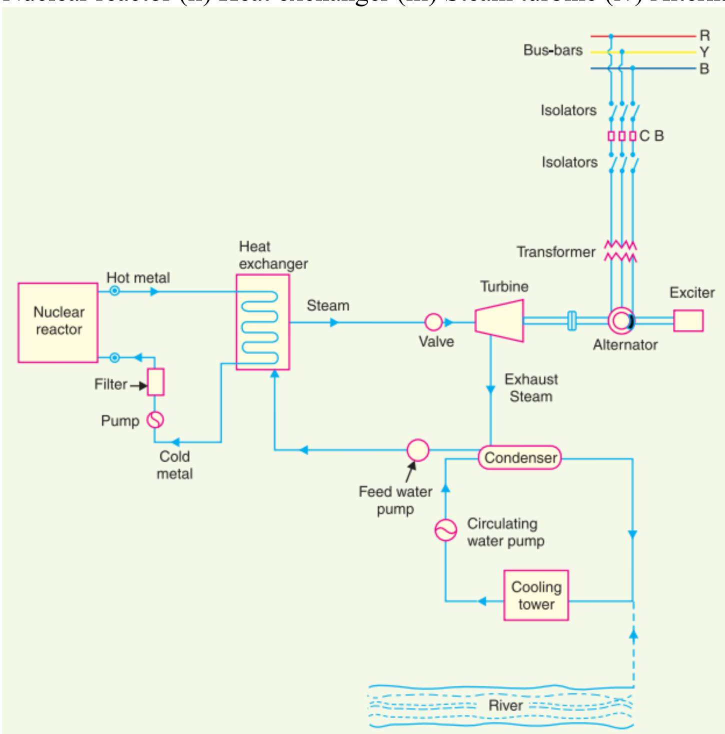
<table>
<tr><td></td><td></td><td>the unit generated being 61.5×106 per annum, calculate (a) demand factor (b) average demand (c) load factor</td></tr>
<tr><td></td><td>ans</td><td>

a) Nuclear power plant

A generating station in which nuclear energy is converted into electrical energy is known as a nuclear power station. In nuclear power station, heavy elements such as Uranium (U^{235}) or Thorium (Th^{232}) are subjected to nuclear fission in a special apparatus known as a reactor. The heat energy thus released is utilised in raising steam at high temperature and pressure. The steam runs the steam turbine which converts steam energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy

The schematic arrangement of a nuclear power station is shown in Fig. The whole arrangement can be divided into the following main stages :

(i) Nuclear reactor (ii) Heat exchanger (iii) Steam turbine (iv) Alternator.



Schematic arrangement of Nuclear Power Station

(i) Nuclear reactor. It is an apparatus in which nuclear fuel (U^{235}) is subjected to nuclear fission. It controls the chain reaction that starts once the fission is done. If the chain reaction is not controlled, the result will be an explosion due to the fast increase in the energy released. A nuclear reactor is a cylindrical stout pressure vessel and houses fuel rods of Uranium, moderator and control rods. The fuel rods constitute the fission material and release huge amount of energy when bombarded with slow moving neutrons. The moderator consists of graphite rods which enclose the fuel rods. The moderator slows down the neutrons before they bombard the fuel rods. The control rods are of cadmium and are inserted into the reactor.

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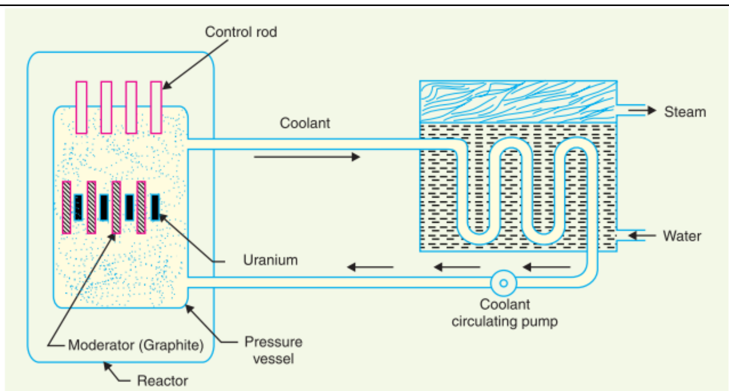


Nuclear Reactor

(ii) Heat exchanger. The coolant gives up heat to the heat exchanger which is utilised in raising the steam. After giving up heat, the coolant is again fed to the reactor.

(iii) Steam turbine. The steam produced in the heat exchanger is led to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted to condenser. The ondenser condenses the steam which is fed to the heat exchanger through feed water pump.

(iv) Alternator. The steam turbine drives the alternator which converts mechanical energy into electrical energy. The output from the alternator is delivered to the bus-bars through transformer, circuit breakers and isolators.

b)

$$\text{Demand factor} = \frac{\text{Maximum demand}}{\text{Connected load}}$$

= 20/23 = 0.8695

$$\text{Yearly average load} = \frac{\text{No. of units (kWh) generated in a year}}{\text{8760 hours}}$$

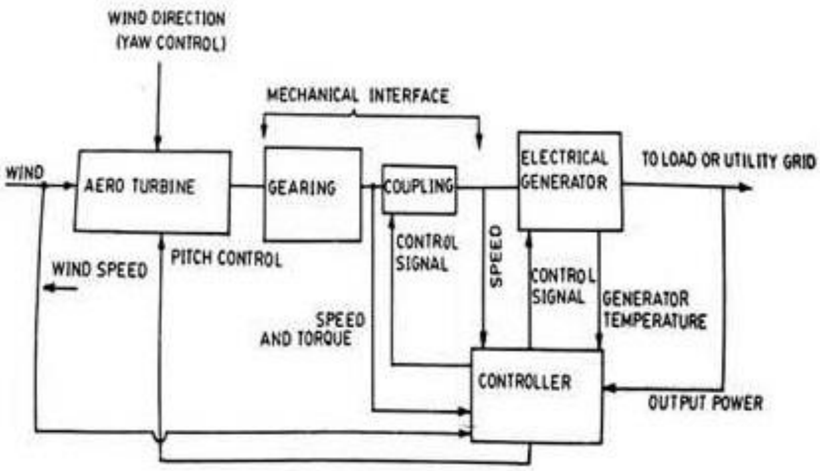
$= (61.5 * 10^6)/(8760) = 7020.55$ KW

$$\text{Load factor} = \frac{\text{Average load}}{\text{Max. demand}}$$

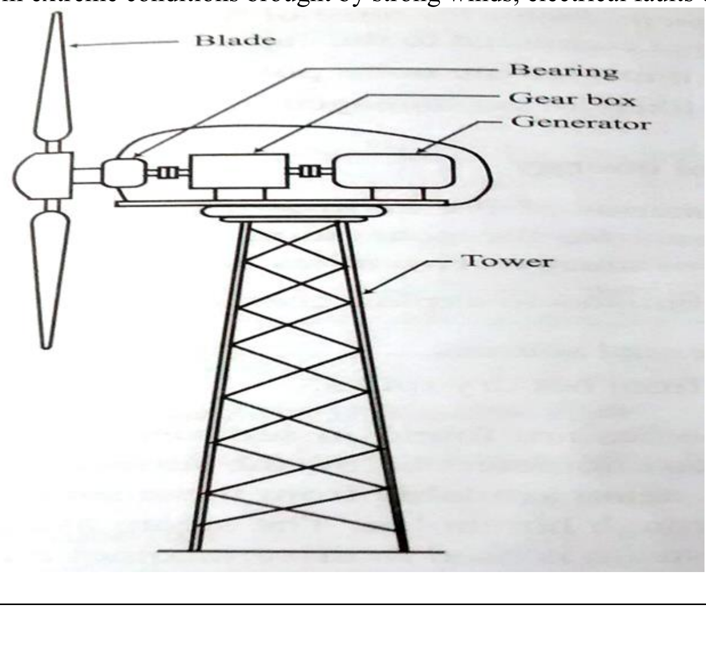
= 7020.55/20,000 = 0.351 = 35.1%

Q12
a) With the help of a block diagram, explain the working of wind energy conversion system.
b) Explain the design steps of a ground mounted solar farm.

Ans
a)
- Airflow can be used to run wind turbines.
- Wind possess energy by virtue of its motion. Any device capable of slowing down the mass of moving air can extract part of the energy and convert it into useful work.
- Wind turbines convert the kinetic energy in the wind into mechanical power.
- A generator can convert mechanical power into electricity.
- A wind electric generator consists of mainly a wind turbine, a generator, a speed changing gear system which couples the turbine with the generator and the control equipments.
- Areas where winds are stronger and more constant, such as offshore and high altitude sites, are preferred locations for wind farms.



- Aero turbines convert energy in moving air to rotary mechanical energy.
- They require pitch and yaw control
- Yaw control: as wind direction changes a motor rotates turbine slowly so as to face the blades into the wind.
- Pitch control: pitch angle is the incident angle of wind with blade. Max power when pitch angle is zero. This control provided by pitch control.
- A mechanical interface consisting of a step up gear and a suitable coupling transmits the rotary mechanical energy to an electric generator.
- Output of this generator connected to load or power grid.
- The purpose of controller is to sense various factors as necessary and appropriate control signals for matching electrical output to wind energy input and protect the system from extreme conditions brought by strong winds, electrical faults etc



b)

1. Select the best spot for ground-mounted solar panels
   - The sunniest spot (full sun)
   - No tree or building shading from 10 am to 3 pm
   - Facing South (in Northern Hemisphere) – facing North (in Northern Hemisphere)
   - Clear the grounds according to the area covered by your solar panels – give yourself an additional 20% of extra space.
2. Design and build your ground structure
   - Concrete foundations
   - GI pipe or wood lumber structure
   - Insecticide wood treatment and outdoor paint/varnish
3. Mount your solar panels on the ground structure
   – Solar panels on aluminum rails
   – Get aluminum mounting brackets and clamps
   – Quick and easy installation (30 min for 6 panels)
4. Connect your solar panels to your inverter
   - Output of solar panels: Positive (+) and Negative (-) wires
   - MC4 male and female solar connectors
   - Solar cable extension to reach the inverter
   - Grounding lug to ground your solar array
5. Maintain your ground-mounted solar panels
   Solar panels are extremely robust. They will last at least 25 years and only require light maintenance such as gentle cleaning from time to time, to keep them at their highest production level. It is also recommended that you inspect your solar frame every six months to assess its structural integrity.

Module 2

Q13	Derive the expression for capacitance of a three phase transmission line with (i) Symmetrical spacing and (ii) Unsymmetrical spacing

ans

- In a 3-phase transmission line, the capacitance of each conductor is considered instead of capacitance from conductor to conductor.
- Here, again two cases arise viz., symmetrical spacing and unsymmetrical spacing
  - **Symmetrical Spacing**
- Fig. 9.24 shows the three conductors A, B and C of the 3-phase overhead transmission line having charges QA, QB and QC per metre length respectively.
- Let the conductors be equidistant (d metres) from each other. We shall find the capacitance from line conductor to neutral in this symmetrically spaced line.
- Referring to Fig., overall potential difference between conductor A and infinite neutral plane is given by

$$V_A = \int_r^\infty \frac{Q_A}{2\pi x\varepsilon_0}\,dx + \int_d^\infty \frac{Q_B}{2\pi x\varepsilon_0}\,dx + \int_d^\infty \frac{Q_C}{2\pi x\varepsilon_0}\,dx$$

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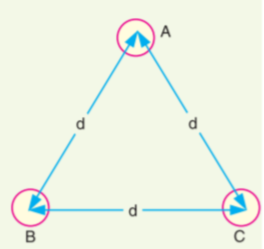


Fig. 9.24

$$= \frac{1}{2\pi\varepsilon_0}\left[Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d} + Q_C \log_e \frac{1}{d}\right]$$

$$= \frac{1}{2\pi\varepsilon_0}\left[Q_A \log_e \frac{1}{r} + (Q_B + Q_C) \log_e \frac{1}{d}\right]$$

Assuming balanced supply, we have, $Q_A + Q_B + Q_C = 0$

$$\therefore \quad Q_B + Q_C = -Q_A$$

$$\therefore \quad V_A = \frac{1}{2\pi\varepsilon_0}\left[Q_A \log_e \frac{1}{r} - Q_A \log_e \frac{1}{d}\right] = \frac{Q_A}{2\pi\varepsilon_0} \log_e \frac{d}{r} \text{ volts}$$

$\therefore$ Capacitance of conductor A w.r.t neutral,

$$C_A = \frac{Q_A}{V_A} = \frac{Q_A}{\frac{Q_A}{2\pi\varepsilon_0} \log_e \frac{d}{r}} \text{ F/m} = \frac{2\pi\varepsilon_0}{\log_e \frac{d}{r}} \text{ F/m}$$

$$\therefore \quad C_A = \frac{2\pi\varepsilon_0}{\log_e \frac{d}{r}} \text{ F/m}$$

- Derived in a similar manner, the expressions for capacitance are the same for conductors B and C.

(ii) Unsymmetrical spacing

- Fig. shows a 3-phase transposed line having unsymmetrical spacing.
- Let us assume balanced conditions i.e. $Q_A + Q_B + Q_C = 0$.
- Considering all the three sections of the transposed line for phase A,

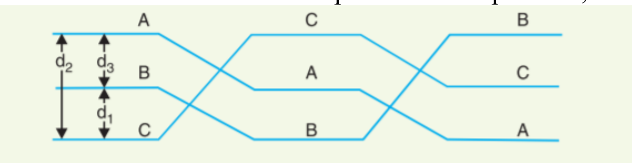


Fig. 9.25

		Potential of 1st position, $V_1 = \frac{1}{2\pi\varepsilon_0}\left(Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_3} + Q_C \log_e \frac{1}{d_2}\right)$ Potential of 2nd position, $V_2 = \frac{1}{2\pi\varepsilon_0}\left(Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_1} + Q_C \log_e \frac{1}{d_3}\right)$ Potential of 3rd position, $V_3 = \frac{1}{2\pi\varepsilon_0}\left(Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_2} + Q_C \log_e \frac{1}{d_1}\right)$ Average voltage on condutor A is $V_A = \frac{1}{3}(V_1 + V_2 + V_3)$ $= \frac{1}{3 \times 2\pi\varepsilon_0}\left[Q_A \log_e \frac{1}{r^3} + (Q_B + Q_C) \log_e \frac{1}{d_1 d_2 d_3}\right]$ As $Q_A + Q_B + Q_C = 0$, therefore, $Q_B + Q_C = -Q_A$ $\therefore \quad V_A = \frac{1}{6\pi\varepsilon_0}\left[Q_A \log_e \frac{1}{r^3} - Q_A \log_e \frac{1}{d_1 d_2 d_3}\right]$ $= \frac{Q_A}{6\pi\varepsilon_0} \log_e \frac{d_1 d_2 d_3}{r^3}$ $= \frac{1}{3} \times \frac{Q_A}{2\pi\varepsilon_0} \log_e \frac{d_1 d_2 d_3}{r^3}$ $= \frac{Q_A}{2\pi\varepsilon_0} \log_e \left(\frac{d_1 d_2 d_3}{r^3}\right)^{1/3}$ $= \frac{Q_A}{2\pi\varepsilon_0} \log_e \frac{(d_1 d_2 d_3)^{1/3}}{r}$ $\therefore$ Capacitance from conductor to neutral is $C_A = \frac{Q_A}{V_A} = \frac{2\pi\varepsilon_0}{\log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r}} F/m$
	Q14	a) What is meant by transposition of lines? What are its advantages. b) In a three phase transmission line with 132KV at the receiving end, the following are the transmission constants: A=D= 0.98<3^o, B=110<75^o Ω, C=0.0005<88^o S. If the load at the receiving end is 50MVA at 0.8 lagging power factor. Determine the voltage, current and power factor at the sending end. c) Derive the expression for inductance of a single phase overhead transmission line.
	Ans	a) When 3-phase line conductors are not equidistant from each other, the conductor spacing is said to be unsymmetrical. Under such conditions, the flux linkages and inductance of each phase are not the same. A different inductance in each phase results in unequal voltage drops in the three phases even if the currents in the conductors are balanced. Therefore, the voltage at the receiving end will not be the

same for all phases. In order that voltage drops are equal in all conductors, we generally interchange the positions of the conductors at regular intervals along the line so that each conductor occupies the original position of every other conductor over an equal distance. Such an exchange of positions is known as transposition. Fig. shows the transposed line. The phase conductors are designated as A, B and C and the positions occupied are numbered 1, 2 and 3. The effect of transposition is that each conductor has the same average inductance.

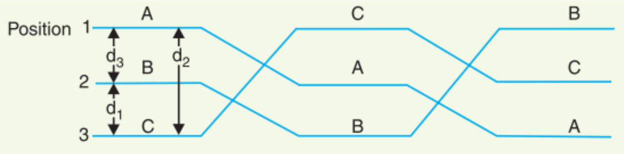


Fig. 9.9

Fig. 9.9 shows a 3-phase transposed line having unsymmetrical spacing. Let us assume that each of the three sections is 1 m in length. Let us further assume balanced conditions *i.e.*, $I_A + I_B + I_C = 0$. Let the line currents be :

$$I_A = I(1 + j\,0)$$
$$I_B = I(-0{\cdot}5 - j\,0{\cdot}866)$$
$$I_C = I(-0{\cdot}5 + j\,0{\cdot}866)$$

b)

(iv) Long lines—Rigorous method. By rigorous method, the sending end voltage and current of a long transmission line are given by :

$$V_S = V_R \cosh\sqrt{Y Z} + I_R\sqrt{\frac{Z}{Y}} \sinh\sqrt{Y Z}$$

$$I_S = V_R\sqrt{\frac{Y}{Z}} \sinh\sqrt{Y Z} + I_R \cosh\sqrt{Y Z}$$

Comparing these equations with those of (*i*) and (*ii*), we get,

$$\vec{A} = \vec{D} = \cosh\sqrt{Y Z}\,; \quad \vec{B} = \sqrt{\frac{Z}{Y}} \sinh\sqrt{Y Z}\,; \quad \vec{C} = \sqrt{\frac{Y}{Z}} \sinh\sqrt{Y Z}$$

Incidentally

$$\vec{A}\vec{D} - \vec{B}\vec{C} = \cosh\sqrt{Y Z} \times \cosh\sqrt{Y Z} - \sqrt{\frac{Z}{Y}} \sinh\sqrt{Y Z} \times \sqrt{\frac{Y}{Z}} \sinh\sqrt{Y Z}$$
$$= \cosh^2\sqrt{Y Z} - \sinh^2\sqrt{Y Z} = 1$$

Sending end voltage, V_s =165.187KV
Sending end current, I_s = 196.88<-24.41°A
Cosø_s=0.8115 Lagging
c)

Consider a single phase overhead line consisting of two parallel conductors A and B spaced d metres apart as shown in Fig. 9.7. Conductors A and B carry the same amount of current (*i.e.* $I_A = I_B$), but in the opposite direction because one forms the return circuit of the other.

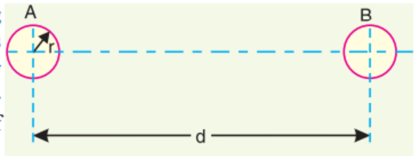


Fig. 9.7

$$\therefore \quad I_A + I_B = 0$$

In order to find the inductance of conductor A (or conductor B), we shall have to consider the flux linkages with it. There will be flux linkages with conductor A due to its own current I_A and also due to the mutual inductance effect of current I_B in the conductor B.

Flux linkages with conductor A due to its own current

$$= \frac{\mu_0 I_A}{2\pi}\left(\frac{1}{4} + \int_r^\infty \frac{dx}{x}\right) \quad \text{...(}i\text{)} \quad \text{[See Art. 9.4]}$$

Flux linkages with conductor A due to current I_B

$$= \frac{\mu_0 I_B}{2\pi}\int_d^\infty \frac{dx}{x} \quad \text{...(}ii\text{)}$$

Total flux linkages with conductor A is

$$\psi_A = \text{exp. } (i) + \text{exp } (ii)$$

$$= \frac{\mu_0 I_A}{2\pi}\left(\frac{1}{4} + \int_r^\infty \frac{dx}{x}\right) + \frac{\mu_0 I_B}{2\pi}\int_d^\infty \frac{dx}{x}$$

$$= \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} + \int_r^\infty \frac{dx}{x}\right) I_A + I_B \int_d^\infty \frac{dx}{x}\right]$$

$$= \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} + \log_e \infty - \log_e r\right) I_A + \left(\log_e \infty - \log_e d\right) I_B\right]$$

$$= \frac{\mu_0}{2\pi}\left[\left(\frac{I_A}{4} + \log_e \infty\,(I_A + I_B) - I_A \log_e r - I_B \log_e d\right)\right]$$

$$= \frac{\mu_0}{2\pi}\left[\frac{I_A}{4} - I_A \log_e r - I_B \log_e d\right] \quad (\because I_A + I_B = 0)$$

Now, $\quad I_A + I_B = 0 \quad \text{or} \quad -I_B = I_A$

$$\therefore \quad -I_B \log_e d = I_A \log_e d$$

$$\therefore \quad \psi_A = \frac{\mu_0}{2\pi}\left[\frac{I_A}{4} + I_A \log_e d - I_A \log_e r\right] \text{ wb-turns/m}$$

$$= \frac{\mu_0}{2\pi}\left[\frac{I_A}{4} + I_A \log_e \frac{d}{r}\right]$$

$$= \frac{\mu_0 I_A}{2\pi}\left[\frac{1}{4} + \log_e \frac{d}{r}\right] \text{ wb-turns/m}$$

Inductance of conductor $A, L_A = \dfrac{\psi_A}{I_A}$

$$= \frac{\mu_0}{2\pi}\left[\frac{1}{4} + \log_e \frac{d}{r}\right] \text{H/m} = \frac{4\pi \times 10^{-7}}{2\pi}\left[\frac{1}{4} + \log_e \frac{d}{r}\right] \text{H/m}$$

$$L_A = 10^{-7}\left[\frac{1}{2} + 2\log_e \frac{d}{r}\right] \text{H/m}$$

$$\text{Loop inductance} = 2\,L_A \text{ H/m} = 10^{-7}\left[1 + 4\log_e \frac{d}{r}\right] \text{H/m}$$

$$\text{Loop inductance} = 10^{-7}\left[1 + 4\log_e \frac{d}{r}\right] \text{H/m}$$

Module 3

Q15 a) What do you mean by sag? Derive the expression for sag when the supports are at same level.

b) What are FACTS devices? How are they classified? Explain the working of any one FACTS device with the help of a diagram.

a) While erecting an overhead line, it is very important that conductors are under safe tension. If the conductors are too much stretched between supports in a bid to save conductor material, the stress in the conductor may reach unsafe value and in certain cases the conductor may break due to excessive tension. In order to permit safe tension in the conductors, they are not fully stretched but are allowed to have a dip or sag.

The difference in level between points of supports and the lowest point on the conductor is called sag.

(i) When supports are at equal levels. Consider a conductor between two equilevel supports *A* and *B* with *O* as the lowest point as shown in Fig. 8.24. It can be proved that lowest point will be at the mid-span.

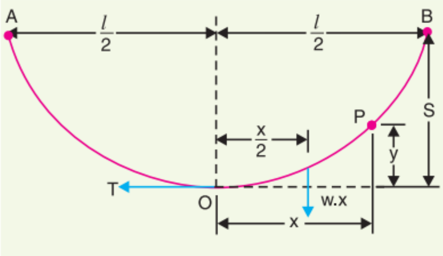


Fig. 8.24

Let

l = Length of span

w = Weight per unit length of conductor

T = Tension in the conductor.

Consider a point *P* on the conductor. Taking the lowest point *O* as the origin, let the co-ordinates of point *P* be *x* and *y*. Assuming that the curvature is so small that curved length is equal to its horizontal projection (*i.e.*, $OP = x$), the two forces acting on the portion *OP* of the conductor are :

(a) The weight wx of conductor acting at a distance $x/2$ from *O*.

(b) The tension *T* acting at *O*.

Equating the moments of above two forces about point *O*, we get,

$$T\,y = w\,x \times \frac{x}{2}$$

or
$$y = \frac{w\,x^2}{2\,T}$$

The maximum dip (sag) is represented by the value of *y* at either of the supports *A* and *B*.

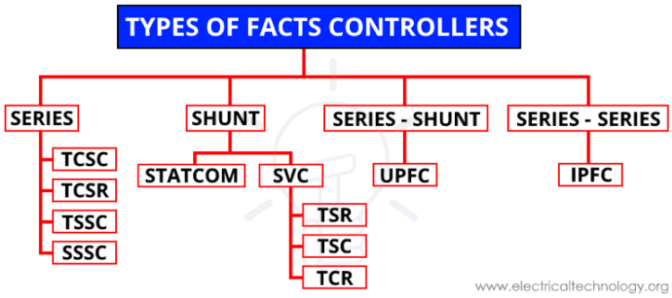
At support *A*, $x = l/2$ and $y = S$

$$\therefore \quad \text{Sag,} \quad S = \frac{w(l/2)^2}{2T} = \frac{w\,l^2}{8\,T}$$

b) FACTS is the acronym for "Flexible AC Transmission Systems". Refers to a group of resources used to overcome certain limitations in the static and dynamic transmission capacity of electrical networks.The main purpose is to supply the network with inductive or capacitive reactive power to its particular requirements, hence improving transmission quality and the efficiency of the power transmission system. The power flow in a transmission line can vary even under normal steady state conditions ie; when load changes. The occurrence of contingency (due to tripping of a line, generator) can result in a sudden increase or decrease in the power flow. This can result in overloading of some lines and consequent threat to system security. The increase in the loading of lines sometimes can lead to voltage collapse due to shortage of reactive power delivered at load centres. This problems are reduced by the introduction of fast dynamic control over reactive & active power by high power electronic controllers.

According to the type of connection FACTS Controller with the power system, it is classified as;

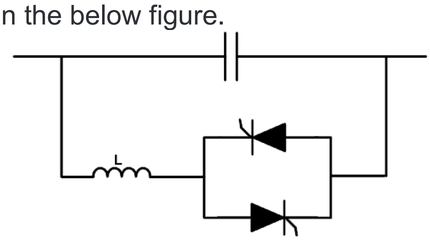
- **Series Connected Controller**
- **Shunt Connected Controller**
- **Combined Series-Series Controller**
- **Combined Shunt-Series Controller**



Thyristor Controlled Series Capacitor (TCSC)

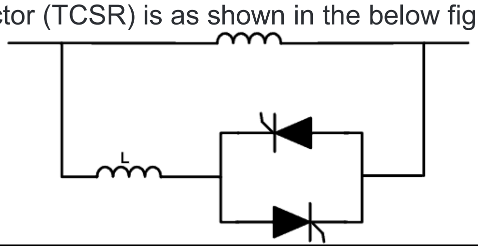
In this method, the capacitive reactance connected in series with the power system. It consists of a capacitor bank in which multiple capacitors are connected in a series-parallel connection. The capacitor bank is connected in parallel with the thyristor-controlled reactor. It is used to provide a smooth variable series capacitance.

The thyristors are used to control the impedance of the system. The total impedance of the circuit can be controlled by controlling the firing angle of a thyristor. A simple block diagram is shown in the below figure.



Thyristor Controlled Series Reactor (TCSR)

This is a series compensator device that provides a smooth variable inductive reactance. This device is the same as TCSC, just the capacitor is replaced with the reactor. The reactor stops conducting when the firing angle is 180˚ and it starts conducting when the firing angle is below 180˚. The basic diagram of Thyristor Controlled Series Reactor (TCSR) is as shown in the below figure.



<table>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td>Q16</td><td>a) A three phase transmission line is being supported by three disc insulators. The potential across top unit and middle unit are 8KV and 11KV respectively. Calculate (i) The ratio of capacitance between pin and earth to the self capacitance of each unit (ii)The line voltage (iii) String efficiency
b) With the help of diagrams, explain what do you mean by intersheath grading.</td></tr>
<tr><td></td><td>Ans</td><td>a)
(i) Let K be the ratio of capacitance between pin and earth to self capacitance. If C farad is the self capacitance of each unit, then capacitance between pin and earth = KC.
Applying Kirchoff's current law to Junction A,
$I_2 = I_1 + i_1$
or $V_2 \omega C = V_1 \omega C + V_1 K \omega C$
or $V_2 = V_1 (1 + K)$
$\therefore \quad K = \frac{V_2 - V_1}{V_1} = \frac{11-8}{8} = \mathbf{0 \cdot 375}$
(ii) Applying Kirchoff's current law to Junction B,
$I_3 = I_2 + i_2$
or $V_3 \omega C = V_2 \omega C + (V_1 + V_2) K \omega C$
or $V_3 = V_2 + (V_1 + V_2) K = 11 + (8 + 11) \times 0 \cdot 375 = 18 \cdot 12 \text{ kV}$
Voltage between line and earth $= V_1 + V_2 + V_3 = 8 + 11 + 18 \cdot 12 = 37 \cdot 12 \text{ kV}$
$\therefore$ Line Voltage $= \sqrt{3} \times 37 \cdot 12 = \mathbf{64 \cdot 28 \text{ kV}}$
(iii) String efficiency $= \frac{\text{Voltage across string}}{\text{No. of insulators} \times V_3} \times 100 = \frac{37 \cdot 12}{3 \times 18 \cdot 12} \times 100 = \mathbf{68.28\%}$
b)
Intersheath Grading
In this method of cable grading, a homogeneous dielectric is used, but it is divided into various layers by placing metallic intersheaths between the core and lead sheath. The intersheaths are held at suitable potentials which are inbetween the core potential and earth potential. This arrangement improves voltage distribution in the dielectric of the cable and consequently more uniform potential gradient is obtained.</td></tr>
</table>

Consider a cable of core diameter d and outer lead sheath of diameter D. Suppose that two intersheaths of diameters d_1 and d_2 are inserted into the homogeneous dielectric and maintained at some fixed potentials. Let V_1, V_2 and V_3 respectively be the voltage between core and intersheath 1, between intersheath 1 and 2 and between intersheath 2 and outer lead sheath. As there is a definite potential difference between the inner and outer layers of each intersheath, therefore, each sheath can be treated like a homogeneous single core cable. As proved in Art. 11.9,

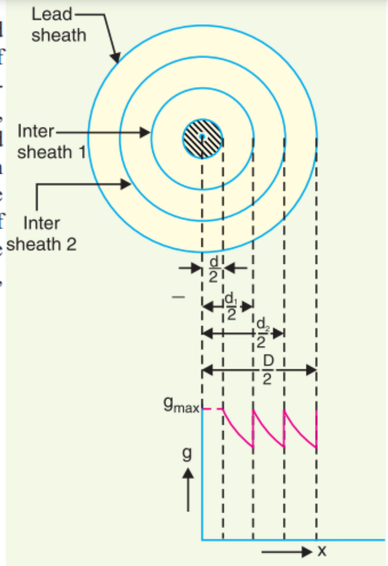


Fig. 11.17

Maximum stress between core and intersheath 1 is

$$g_{1max} = \frac{V_1}{\frac{d}{2}\log_e \frac{d_1}{d}}$$

Similarly,

$$g_{2max} = \frac{V_2}{\frac{d_1}{2}\log_e \frac{d_2}{d_1}}$$

$$g_{3max} = \frac{V_3}{\frac{d_2}{2}\log_e \frac{D}{d_2}}$$

Since the dielectric is homogeneous, the maximum stress in each layer is the same *i.e.,*

$$g_{1max} = g_{2max} = g_{3max} = g_{max} \text{ (say)}$$

$$\therefore \quad \frac{V_1}{\frac{d}{2}\log_e \frac{d_1}{d}} = \frac{V_2}{\frac{d_1}{2}\log_e \frac{d_2}{d_1}} = \frac{V_3}{\frac{d_2}{2}\log_e \frac{D}{d_2}}$$

As the cable behaves like three capacitors in series, therefore, all the potentials are in phase *i.e.*

Voltage between conductor and earthed lead sheath is

$$V = V_1 + V_2 + V_3$$

Module 4

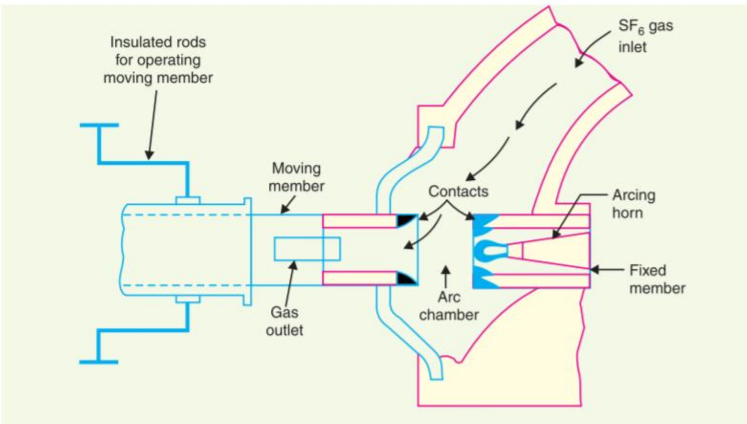
Q17 a) What are the basic requirements of a protective relaying?
b) Explain the construction and working of SF6 Circuit breaker.

Ans a) Protective relay system should have the following qualities:

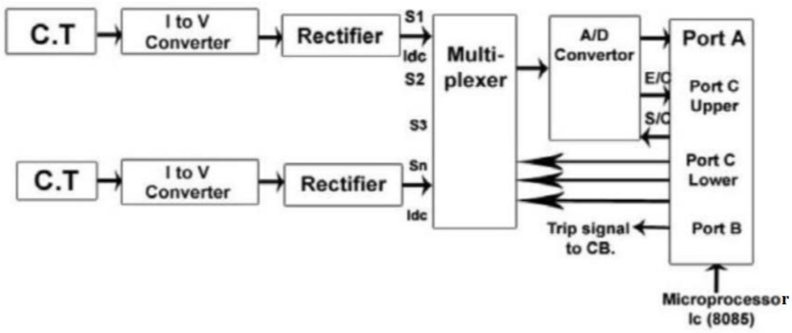
1. **Selectivity**: It is the ability of the protective system to select correctly that part of the system in trouble and disconnect the faulty part without disturbing the rest of the system. For that dividing the entire system into several protective zones.
2. **Speed:** The relay system should disconnect the faulty section as fast as possible.
3. **Sensitivity:** It is the ability of the relay system to operate with low value of actuating quantity.
4. **Reliability** It is the ability of the relay system to operate under the pre-determined conditions. Without reliability, the protection would be rendered largely ineffective and could even become a liability.
5. **Simplicity**. The relaying system should be simple so that it can be easily maintained. Reliability is closely related to simplicity. The simpler the protection scheme, the greater will be its reliability.
6. **Economy.** The most important factor in the choice of a particular protection scheme is the economic aspect.

b) **SF_6 Circuit Breaker**

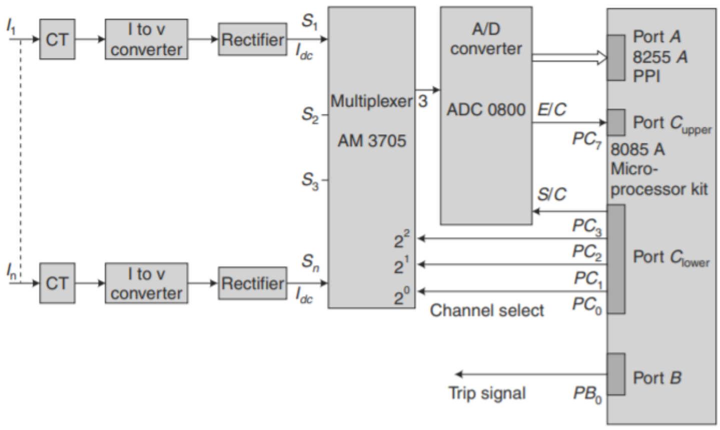
- In such circuit breakers, sulphur hexaflouride (SF_6) gas is used as the arc quenching medium.

		➢ The SF_6 is an electro-negative gas and has a strong tendency to absorb free electrons. ➢ The contacts of the breaker are opened in a high pressure flow of SF_6 gas and an arc is struck between them. ➢ The conducting free electrons in the arc are rapidly captured by the gas to form relatively immobile negative ions. ➢ This loss of conducting electrons in the arc quickly builds up enough insulation strength to extinguish the arc. ➢ The SF_6 circuit breakers have been found to be very effective for high power and high voltage service. ➢ It consists of fixed and moving contacts enclosed in a chamber (called arc interruption chamber) containing SF_6 gas. ➢ This chamber is connected to SF_6 gas reservoir. ➢ When the contacts of breaker are opened, the valve mechanism permits a high pressure SF_6 gas from the reservoir to flow towards the arc interruption chamber. ➢ The fixed contact is a hollow cylindrical current carrying contact fitted with an arc horn. ➢ The moving contact is also a hollow cylinder with rectangular holes in the sides to permit the SF_6 gas to let out through these holes after flowing along and across the arc. ➢ The tips of fixed contact, moving contact and arcing horn are coated with copper-tungsten arc resistant material. ➢ Since SF_6 gas is costly, it is reconditioned and reclaimed by suitable auxiliary system after each operation of the breaker. ➢ In the closed position of the breaker, the contacts remain surrounded by SF_6 gas at a pressure of about 2·8 kg/cm^2. ➢ When the breaker operates, the moving contact is pulled apart and an arc is struck between the contacts. ➢ The movement of the moving contact is synchronized with the opening of a valve which permits SF_6 gas at 14 kg/cm^2 pressure from the reservoir to the arc interruption chamber. ➢ The high pressure flow of SF_6 rapidly absorbs the free electrons in the arc path to form immobile negative ions which are ineffective as charge carriers. ➢ The result is that the medium between the contacts quickly builds up high dielectric strength and causes the extinction of the arc. ➢ After the breaker operation (i.e., after arc extinction), the valve is closed by the action of a set of springs. 
	Q18	Explain the working of microprocessor based over current relay with the help of block diagram.

		b) Explain with the help of a diagram, the construction and working of any one type of surge diverter.
	Ans	**a) Microprocessor based over current relay** ➢ An over current relay is the simplest form of protective relay which operates when the current in any circuit exceeds a certain predetermined value ie the pick up value. ➢ Using a multiplexer the microprocessor can sense the fault current of a number of circuits. ➢ If the fault current in any circuit exceeds pickup value the microprocessor sends a tripping signal to the circuit breaker of the faulty circuit. ➢ Micro processor accepts signal in voltage form. ➢ So current is taken from C.T. and given to I to V converter because many electronics circuit require voltage signal for operation. ➢ The A.C. voltage is converted into D.C. voltage by using rectifier. ➢ This D.C. voltage is proportional to load current only. The output of rectifier is given to Multiplexer. ➢ The Multiplexer gives output to A/D Converter where Analog DC voltage is converted to Digital form (in form of 0 and 1 i.e. binary form). Microprocessor understands only codes in 0 and 1 form. ➢ μP (Microprocessor) gives S/C (start of conversion) signal to A/D converter (i.e. analog to digital conversion is started and μ P gives permission to A/D convertor for this by sending S/C) When converting from analog to digital is over (finish) then A/D converter sends E/C signal to μP (E/C – End of Conversion). ➢ When work of A/D is over then μP compare the magnitude of this incoming current with required current value (i.e. set value or reference value). ➢ If incoming value is more – fault is occurring and trip signal is sent to CB (Circuit Breaker). ➢ An overcurrent relay is the simplest form of protective relay which operates when the current in any circuit exceeds a certain predetermined value, i.e. the pick-up value. ➢ It is extensively used for the protection of distribution lines, industrial motors and equipment. ➢ Using a multiplexer, the microprocessor can sense the fault currents of a number of circuits. ➢ If the fault current in any circuit exceeds the pick-up value, the microprocessor sends a tripping signal to the circuit breaker of the faulty circuit breaker of the faulty circuit. ➢ As the microprocessor accepts signals in voltage form, the current signal derived from the current transformer is converted into a proportional voltage signal using a current to voltage converter. ➢ The ac voltage proportional to the load current is converted into dc using a precision rectifier. Thus, the microprocessor accepts dc voltage proportional to load current. ➢ The block schematic diagram of the relay is shown in Figure. ➢ The output of the rectifier is fed to the multiplexer. ➢ The microcomputer sends a command to switch on the desired channel of the multiplexer to obtain the rectified voltage proportional to the current in a particular circuit. ➢ The output of the multiplexer is fed to the A/D converter to obtain the signal in digital form. The A/D converter ADC0800 has been used for this purpose. The microcomputer sends a signal to the ADC for starting the conversion. ➢ The microcomputer reads the end of conversion signal to examine whether the conversion is over or not. ➢ As soon as the conversion is over, the microcomputer reads the current signal in digital form and then compares it with the pick-up value.



Microprocessor Based Over Current Relay



b) Surge Diverter

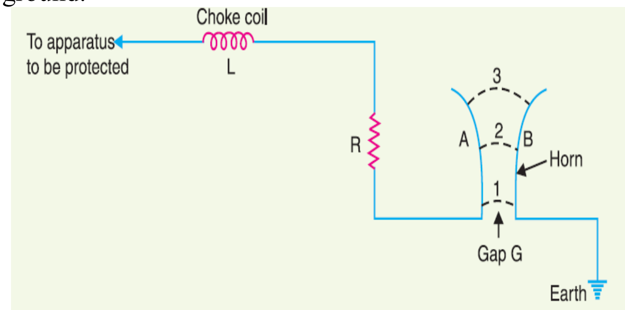
A lightning arrester or a surge diverter is a protective device which conducts the high voltage surges on the power system to the ground, thus protecting sensitive electrical and electronic equipment.

It consists of a spark gap in series with a non-linear resistor. One end of the diverter is connected to the terminal of the equipment to be protected and the other end is effectively grounded. The length of the gap is so set that normal line voltage is not enough to cause an arc across the gap but a dangerously high voltage will break down the air insulation and form an arc. The property of the non-linear resistance is that its resistance decreases as the voltage (or current) increases and vice-versa.

HORN GAP ARRESTOR

- It consists of two horn shaped metal rods A and B separated by a small air gap. The horns are so constructed that distance between them gradually increases towards the top as shown.
- One end of horn is connected to the line through a resistance R and choke coil L while the other end is effectively grounded. The resistance R helps in limiting the follow current to a small value.
- The choke coil is so designed that it offers small reactance at normal power frequency but a very high reactance at transient frequency. Thus the choke does not allow the transients to enter the apparatus to be protected. The gap between the horns is so adjusted that normal supply voltage is not enough to cause an arc across the gap.

		• Under normal conditions, the gap is non-conducting i.e. normal supply voltage is insufficient to initiate the arc between the gap. On the occurrence of an overvoltage, spark-over takes place across the small gap G. The heated air around the arc and the magnetic effect of the arc cause the arc to travel up the gap. The arc moves progressively into positions 1, 2 and 3. At some position of the arc (perhaps position 3), the distance may be too great for the voltage to maintain the arc. Consequently, the arc is extinguished. The excess charge on the line is thus conducted through the arrester to the ground.



Module 5

	Q19	a) Derive an expression for the most economical value of power factor which may be attained by a consumer. b) A single phase distributor 2 kilometres long supplies a load of 120 A at 0·8 p.f. lagging at its far end and a load of 80 A at 0·9 p.f. lagging at its mid-point. Both power factors are referred to the voltage at the far end. The resistance and reactance per km are 0·05 Ω and 0·1 Ω respectively. If the voltage at the far end is maintained at 230 V, calculate: (i) voltage at the sending end (ii) phase angle between voltages at the two ends
	Ans	a) If a consumer improves the power factor, there is reduction in his maximum kVA demand and hence there will be annual saving over the maximum demand charges. However, when power factor is improved, it involves capital investment on the power factor correction equipment. The consumer will incur expenditure every year in the shape of annual interest and depreciation on the investment made over the p.f. correction equipment. Therefore, the net annual saving will be equal to the annual saving in maximum demand charges minus annual expenditure incurred on p.f. correction equipment. ➢ The value to which the power factor should be improved so as to have maximum net annual saving is known as the most economical power factor. ➢ Consider a consumer taking a peak load of P kW at a power factor of cos φ_1 and charged at a rate of Rs x per kVA of maximum demand per annum. Suppose the consumer improves the power factor

to cos ϕ_2 by installing p.f. correction equipment. Let expenditure incurred on the p.f. correction equipment be Rs y per kVAR per annum. The power triangle at the original p.f. cos ϕ_1 is OAB and for the improved p.f. cos ϕ_2, it is OAC [See Fig. 6.13].

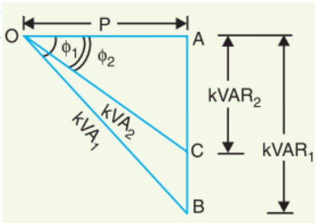


Fig. 6.13

kVA max. demand at cos ϕ_1, $kVA_1 = P/\cos\phi_1 = P\sec\phi_1$

kVA max. demand at cos ϕ_2, $kVA_2 = P/\cos\phi_2 = P\sec\phi_2$

Annual saving in maximum demand charges

$$= \text{Rs } x\,(kVA_1 - kVA_2)$$
$$= \text{Rs } x\,(P\sec\phi_1 - P\sec\phi_2)$$
$$= \text{Rs } x\,P\,(\sec\phi_1 - \sec\phi_2) \quad \text{...}(i)$$

Reactive power at cos ϕ_1, $kVAR_1 = P\tan\phi_1$

Reactive power at cos ϕ_2, $kVAR_2 = P\tan\phi_2$

Leading kVAR taken by p.f. correction equipment

$$= P\,(\tan\phi_1 - \tan\phi_2)$$

Annual cost of p.f. correction equipment

$$= \text{Rs } Py\,(\tan\phi_1 - \tan\phi_2) \quad \text{...}(ii)$$

$$\text{Net annual saving, } S = \text{exp. }(i) - \text{exp. }(ii)$$
$$= xP\,(\sec\phi_1 - \sec\phi_2) - yP\,(\tan\phi_1 - \tan\phi_2)$$

In this expression, only ϕ_2 is variable while all other quantities are fixed. Therefore, the net annual saving will be maximum if differentiation of above expression *w.r.t.* ϕ_2 is zero *i.e.*

$$\frac{d}{d\phi_2}(S) = 0$$

or $$\frac{d}{d\phi_2}[xP\,(\sec\phi_1 - \sec\phi_2) - yP\,(\tan\phi_1 - \tan\phi_2)] = 0$$

or $$\frac{d}{d\phi_2}(xP\sec\phi_1) - \frac{d}{d\phi_2}(xP\sec\phi_2) - \frac{d}{d\phi_2}(yP\tan\phi_1) + yP\frac{d}{d\phi_2}(\tan\phi_2) = 0$$

or $$0 - xP\sec\phi_2\tan\phi_2 - 0 + yP\sec^2\phi_2 = 0$$

or $$-x\tan\phi_2 + y\sec\phi_2 = 0$$

or $$\tan\phi_2 = \frac{y}{x}\sec\phi_2$$

or $$\sin\phi_2 = y/x$$

$\therefore$ Most economical power factor, $\cos\phi_2 = \sqrt{1-\sin^2\phi_2} = \sqrt{1-(y/x)^2}$

It may be noted that the most economical power factor (cos ϕ_2) depends upon the relative costs of supply and p.f. correction equipment but is independent of the original p.f. cos ϕ_1.

b)

Impedance of distributor/km $= (0{\cdot}05 + j\,0{\cdot}1)\,\Omega$

Impedance of section AC, $\vec{Z}_{AC} = (0{\cdot}05 + j\,0{\cdot}1) \times 1000/1000 = (0{\cdot}05 + j\,0{\cdot}1)\,\Omega$

Impedance of section CB, $\vec{Z}_{CB} = (0{\cdot}05 + j\,0{\cdot}1) \times 1000/1000 = (0{\cdot}05 + j\,0{\cdot}1)\,\Omega$

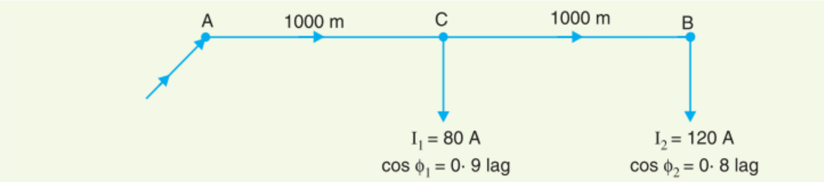


Fig. 14.5

Let the voltage V_B at point B be taken as the reference vector.

Then, $\vec{V_B} = 230 + j\,0$

(*i*) Load current at point B, $\vec{I_2} = 120\,(0{\cdot}8 - j\,0{\cdot}6) = 96 - j\,72$

Load current at point C, $\vec{I_1} = 80\,(0{\cdot}9 - j\,0{\cdot}436) = 72 - j\,34{\cdot}88$

Current in section CB, $\vec{I_{CB}} = \vec{I_2} = 96 - j\,72$

Current in section AC, $\vec{I_{AC}} = \vec{I_1} + \vec{I_2} = (72 - j\,34{\cdot}88) + (96 - j\,72)$

$= 168 - j\,106{\cdot}88$

Drop in section CB, $\vec{V_{CB}} = \vec{I_{CB}}\,\vec{Z_{CB}} = (96 - j\,72)\,(0{\cdot}05 + j\,0{\cdot}1)$

$= 12 + j\,6$

Drop in section AC, $\vec{V_{AC}} = \vec{I_{AC}}\,\vec{Z_{AC}} = (168 - j\,106{\cdot}88)\,(0{\cdot}05 + j\,0{\cdot}1)$

$= 19{\cdot}08 + j\,11{\cdot}45$

$\therefore$ Sending end voltage, $\vec{V_A} = \vec{V_B} + \vec{V_{CB}} + \vec{V_{AC}}$

$= (230 + j\,0) + (12 + j\,6) + (19.08 + j\,11.45)$

$= 261.08 + j\,17.45$

Its magnitude is $= \sqrt{(261{\cdot}08)^2 + (17{\cdot}45)^2} =$ **261·67 V**

(*ii*) The phase difference θ between V_A and V_B is given by :

$$\tan\theta = \frac{17{\cdot}45}{261{\cdot}08} = 0{\cdot}0668$$

$$\therefore \quad \theta = \tan^{-1} 0{\cdot}0668 = \mathbf{3{\cdot}82^{o}}$$

Q20 a) Write short notes on the following

i Two-part tariff

ii Three part tariff

iii Power factor tariff

b) Write short notes on distribution automation system.

c) What do you mean by an aerial bunched cable? Compare its advantages and disadvantages.

Ans a) The rate at which electrical energy is supplied to a consumer is known as tariff.

Two-part tariff - When the rate of electrical energy is charged on the basis of maximum demand of the consumer and the units consumed, it is called a two-part tariff. In two-part tariff, the total charge to be made from the consumer is split into two components viz., fixed charges and running charges. The fixed charges depend upon the maximum demand of the consumer while the running charges depend upon the number of units consumed by the consumer. Thus, the consumer is charged at a certain amount per kW of maximum demand plus a certain amount per kWh of energy consumed i.e.,

$$\text{Total charges} = \text{Rs}\,(b \times \text{kW} + c \times \text{kWh})$$

where, b = charge per kW of maximum demand

c = charge per kWh of energy consumed

This type of tariff is mostly applicable to industrial consumers who have appreciable maximum demand.

Power factor tariff - The tariff in which power factor of the consumer's load is taken into consideration is known as power factor tariff.

In an a.c. system, power factor plays an important role. A low* power factor increases the rating of station equipment and line losses. Therefore, a consumer having low power factor must be penalised. The following are the important types of power factor tariff :

(i) *k VA maximum demand tariff* : It is a modified form of two-part tariff. In this case, the fixed charges are made on the basis of maximum demand in kVA and *not* in kW. As kVA is inversely proportional to power factor, therefore, a consumer having low power factor has to contribute more towards the fixed charges. This type of tariff has the advantage that it encourages the consumers to operate their appliances and machinery at improved power factor.

(ii) *Sliding scale tariff* : This is also know as average power factor tariff. In this case, an average power factor, say 0·8 lagging, is taken as the reference. If the power factor of the consumer falls below this factor, suitable additional charges are made. On the other hand, if the power factor is above the reference, a discount is allowed to the consumer.

(iii) *kW and kVAR tariff* : In this type, both active power (kW) and reactive power (kVAR) supplied are charged separately. A consumer having low power factor will draw more reactive power and hence shall have to pay more charges.

Three-part tariff - When the total charge to be made from the consumer is split into three parts viz., fixed charge, semi-fixed charge and running charge, it is known as a three-part tariff. i.e.,

$$\text{Total charge} = \text{Rs}\,(a + b \times \text{kW} + c \times \text{kWh})$$

a = fixed charge made during each billing period. It includes interest and depreciation on the cost of secondary distribution and labour cost of collecting revenues,

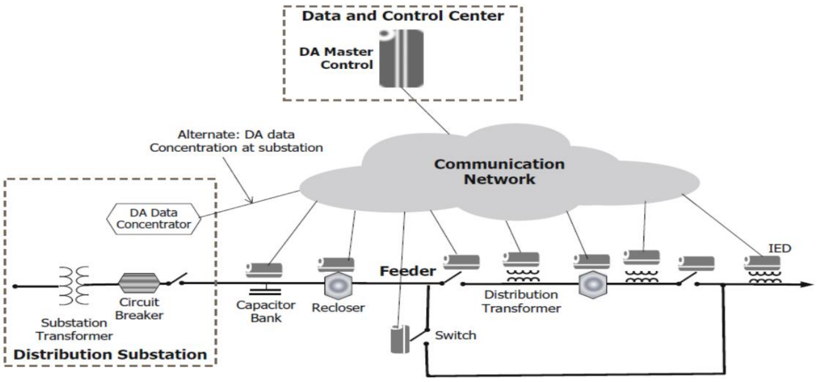
b = charge per kW of maximum demand,

c = charge per kWh of energy consumed.

It may be seen that by adding fixed charge or consumer's charge (i.e., a) to two-part tariff, it becomes three-part tariff. The principal objection of this type of tariff is that the charges are split into three components. This type of tariff is generally applied to big consumers.

b) Distribution Automation

- Distribution automation refers to the automation of all functions related to the distribution system using information collected from substation devices, devices deployed on feeders, and meters deployed at consumer locations.
- Thus, SCADA system that monitors and controls distribution substations is considered a DA function. More widely, definition of distribution automation is limited to the acquisition of data (measurements) from IEDs connected to the devices on the feeder and the control of those feeder devices.



1. Reclosers:

• A recloser monitors and breaks the feeder circuit if the current exceeds a preset threshold. Attempts to reconnect the circuit automatically after a short period of time after disconnecting it. Attempts to reconnect several preconfigured times before "concluding" that the fault is permanent. In the event a recloser declares a fault permanent, reclosers must be operated manually via a remotely executed command.

2. Switches:

• Switches on feeders are deployed to "sectionalize" faulty sections of the feeders and to divert power around a faulty section (until the fault is repaired). Switches are operated manually or by using remotely executed commands.

- Each of these devices must include IEDs to support the required measurement, monitoring, and control functions. These IEDs communicate with the DA control system, called the DA master control. A DA data concentrator collects data from IEDs on one or more feeders and forwards the data to the DA master.

c) Aerial bunched cable

- Aerial bundled cables (also aerial bundled conductors or simply ABC) are overhead power lines using several insulated phase conductors bundled tightly together, usually with a bare neutral conductor.
- This contrasts with the traditional practice of using uninsulated conductors separated by air gaps.
- This variation of bundled conductors utilizes the same principles as overhead power lines, except that they are closer together to the point of touching but each conductor is surrounded by an insulating layer (except for the neutral line).
- The main objections to the traditional design are that the multiple conductors are considered unappealing, and external forces (such as high winds) can cause them to touch and short circuit.

Advantages

- Relative immunity to short circuits caused by external forces (wind, fallen branches), unless they abrade the insulation.
- Can stand in close proximity to trees/buildings and will not generate sparks if touched.
- Little to no tree trimming necessary
- Simpler installation, as crossbars and insulators are not required.
- More aesthetically appealing.
- Significantly improved safety for linespersons, particularly when working on live conductors.
- Electricity theft is made harder and more obvious to detect.
- Less required maintenance and necessary inspections of lines.
- Improved reliability in comparison with both bare conductor overhead systems and underground systems. Insulated conductors prevent accidental contact and supply can be maintained temporarily in the event of a suspension system collapse.

Disadvantages

- Additional cost for the cable itself.
- Insulation degrades due to sun exposure, though the critical insulation between the wires is somewhat shielded from the sun.
- Shorter spans and more poles due to increased weight.
- Can lead to much longer repair times for installations in hilly areas due to much higher line weights requiring bigger and more specialized equipment to repair.
- Older installations are known to cause fires in areas where falling large trees or branches regularly cause breaks in lines and or in insulation leading to short circuits which can then lead to burning insulation dripping to ground and starting ground fires.